Detailed Storage Roadmap for AI-Powered IDE Project

1. Database Fundamentals

Why We Use It: Database fundamentals help understand how data is stored, managed, queried, and optimized in applications.

Topics to Learn Deeply:
• SQL vs NoSQL
• CRUD operations
• Relationships
• Transactions
• Indexing
• Query optimization
• Data modeling

Best Source: https://www.geeksforgeeks.org/dbms/

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2. MongoDB

Why We Use It: MongoDB stores user projects, AI chats, settings, logs, and JSON-based application data.

Topics to Learn Deeply:
• Collections and documents
• CRUD operations
• Aggregation
• Indexes
• Schema design
• MongoDB Atlas
• Performance optimization

Best Source: https://www.mongodb.com/docs/

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3. PostgreSQL

Why We Use It: PostgreSQL is used for structured and relational data like billing, analytics, and authentication systems.

Topics to Learn Deeply:
• Tables
• Joins
• Normalization
• Primary/foreign keys
• Transactions
• Advanced queries
• Indexes

Best Source: https://www.postgresql.org/docs/

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4. Redis

Why We Use It: Redis improves performance using caching and supports sessions, rate limiting, and real-time systems.

Topics to Learn Deeply:
• Caching
• Session storage
• Expiry
• Pub/Sub
• Queues
• Memory optimization

Best Source: https://redis.io/docs/

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5. File Storage System

Why We Use It: File storage systems manage user code files, uploads, execution outputs, and project directories.

Topics to Learn Deeply:
• File handling
• Directory structures
• Uploads/downloads
• Temporary files
• Compression
• Binary storage

Best Source: https://docs.python.org/3/tutorial/inputoutput.html

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6. Cloud Storage

Why We Use It: Cloud storage helps store files online securely and supports scalability and backups.

Topics to Learn Deeply:
• AWS S3
• Firebase Storage
• Bucket management
• Upload APIs
• CDN basics
• Signed URLs

Best Source: https://docs.aws.amazon.com/s3/

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7. Vector Database

Why We Use It: Vector databases are used for AI memory, semantic search, embeddings, and RAG systems.

Topics to Learn Deeply:
• Embeddings
• Similarity search
• Vector indexing
• Metadata filtering
• RAG architecture

Best Source: https://docs.pinecone.io/

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8. ORM / ODM

Why We Use It: ORM and ODM tools simplify database interaction using programming language objects instead of raw queries.

Topics to Learn Deeply:
• SQLAlchemy
• Mongoose
• Prisma
• Beanie
• Model relationships
• Database migrations

Best Source: https://www.sqlalchemy.org/

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9. Database Security

Why We Use It: Database security protects sensitive user information, authentication data, and AI conversations.

Topics to Learn Deeply:
• Encryption
• Hashing
• Access control
• Injection prevention
• Backups
• Credential management

Best Source: https://owasp.org/www-project-top-ten/

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10. Backup & Scalability

Why We Use It: Backup and scalability techniques help prevent data loss and support growing user traffic.

Topics to Learn Deeply:
• Replication
• Sharding
• Auto backups
• Horizontal scaling
• Database optimization

Best Source: https://www.mongodb.com/docs/manual/sharding/